CALTECH SURF RESEARCH PROPOSAL (SUMMER 2027)

OptiBox: Experimental Verification and Multi-Channel Characterization of Phase-Change Photonic Tensor Accelerators for Edge Generative AI

Dalton Omondi
Department of Computational Chemistry and Mathematical Physics
Nairobi, Kenya
Email: daltonomondi588@gmail.com
Prospective Caltech Faculty Mentors: Prof. Alireza Marandi / Prof. Kerry Vahala

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Abstract

Autoregressive large language model (LLM) inference at batch size $B = 1$ is severely throttled by the electronic memory wall: moving weights across copper wires consumes $> 90\%$ of total accelerator energy. While all-optical weight storage suffers from an impossible chip-area scaling penalty (> 2.5 m^2 silicon for 4B parameters), our work develops **OptiBox**: a hybrid, tiered architecture pairing bulk LPDDR5X DRAM with a compact non-volatile optical phase-change memory (OPCM) weight cache ($Ge_2Sb_2Se_4Te_1$) and a high-bandwidth silicon photonic tensor core (20 GHz WDM). In this 10-week Caltech SURF project, we propose to empirically validate the OptiBox break-even amortization model, measure RF/optical SNR and Effective Number of Bits (ENOB) on Caltech nanophotonics testbenches, and prototype the open-source GDSFactory layout targeting imec/AIM multi-project wafer (MPW) tape-outs.

1 Introduction & Background

The exponential expansion of generative AI models (4B to 70B parameters) has created an unsustainable energy and thermal crisis. Digital CMOS accelerators (e.g., NVIDIA H100/H200, Blackwell) dissipate 400–700 W, requiring loud fans or liquid chillers. For edge appliances—such as a compact (25 cm^3), quiet desktop assistant powered by USB-C (≤ 35 W)—CMOS memory buses impose an impassable physical limit (≈ 2.5 pJ/bit DRAM transfers).

Photonic integrated circuits (PICs) offer sub-femtosecond analog propagation and ultra-wide wavelength-division multiplexing (WDM) bandwidth. However, recent literature reveals two fatal traps in naive optical acceleration:

1. **The Area Trap**: Placing all model weights into optical waveguides requires > 35 silicon wafers for a single 70B model.

2. **The Precision and Conversion Tax**: DACs, ADCs, and laser wall-plug efficiency rapidly dominate unless amortized over repeated matrix projections.

To resolve these bottlenecks, we formulated the **OptiBox Heterogeneous Tiered Architecture**, proving a rigorous closed-form break-even horizon:

$$H^* = \frac{\frac{8}{b_{\text{cell}}} E_{\text{pcm_prog}}}{E_{\text{dram_read}} - E_{\text{opt_read}} + \Delta E_{\text{compute}}} \approx 48\text{–}54 \text{ tokens} \tag{1}$$

Beyond 54 tokens, OptiBox delivers 35%–82% lower energy per token and up to 4.7× speedup over digital CMOS within a strict 14.2–22.6 W power budget.

2 Research Objectives & 10-Week SURF Work Plan

Under the mentorship of the Caltech Nanophotonics and Ultrafast Optics laboratories (Prof. Alireza Marandi / Prof. Kerry Vahala), this SURF project aims to transition the theoretical and trace-driven OptiBox framework into physical laboratory reality.

Table 1: Caltech SURF 10-Week Research Milestones

Weeks	Objective	Deliverable & Methodology
Weeks 1–2	Optical Comb & Modulator Calibration	Characterize Kerr soliton microcomb noise (RIN ≤ -150 dBc/Hz) and 20 GHz electro-optic modulators.
Weeks 3–4	OPCM Multi-Level Switching Physics	Test Sb_2Se_3 / GSSe phase-change cells on silicon waveguides; measure optical insertion loss (< 0.8 dB) and cycling endurance ($> 5 \times 10^5$).
Weeks 5–6	High-Speed RF & SNR/ENOB Benchmarking	Measure analog optical accumulation using high-speed SiGe photodetectors and 20 GS/s digitizers; verify ENOB ≥ 6.0 bits.
Weeks 7–8	End-to-End LLM Trace Acceleration	Stream LLaMA-3 (8B) and Qwen (4B) activation traces through the optical crossbar testbench; validate INT4 inference fidelity.
Weeks 9–10	MPW Tape-Out Layout & Final Report	Complete GDSFactory layout for imec iSiPP50G MPW run; publish Caltech SURF report and submit to OFC/CLEO.

3 Methodology & Open-Source Infrastructure

Rather than relying on closed proprietary toolchains (Cadence Virtuoso, Lumerical), our work builds on fully open-source and scriptable Python infrastructure:

- **GDSFactory**: Scripted parametric layout for waveguides, directional couplers, and thermo-optic phase shifters mapped to imec iSiPP50G PDK rules.
- **MIT Meep / SAX**: Differentiable Maxwell solver and scattering-matrix circuit simulations to model modal dispersion and crosstalk.
- **OptiBox Simulator**: Validated trace-driven engine modeling LPDDR5X DRAM, DAC/ADC quantization noise, and Arrhenius thermal stability.

4 Expected Impact & Alignment with Caltech's Mission

Caltech has pioneered high-Q microcavities, optical frequency combs, and ultrafast nonlinear photonic computing. This project directly leverages Caltech's unique strengths to resolve the generative AI hardware crisis. Proving that a desk-side photonic AI appliance can operate silently under 35 W with a $>$ 15-year operational lifespan creates an immediate pathway for both academic breakthroughs and commercial translation.

References

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